

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			817	USE ONLY FOR TIE IN
20										821	TO BE USED FOR LINES SUBJECT TO AIV OR WHERE REQUIRED BY STRESS.
20	U04		MODULAR DBB	K06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
30	U04	VAW	MODULAR DBB	K06AC06	2	3	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
40	U04		MODULAR DBB	M06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
50	U04		MODULAR DBB	M06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	BA6	FOR TURN AROUND TIE-INS ONLY
60	U04	VAW	MODULAR DBB	M06EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	3	FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
70	U04	VAQ	MODULAR DBB	MC5C4	1	1	MODULAR VALVE DB&B GATE TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
80		VB2					***** BALL VALVES *****			NT5	TO BE USED IN NON-PIGGED LINES ONLY
80										SB6	USE JACKSCREWS FOR PADDLE
90											
90	U04		BALL	A06AC05	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB LO	0	0		
100	U04		BALL	A06AC06	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB LO	0	0		
110	U04		BALL	A06AC07	6	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0		
120	U04		BALL	A06AC08	6	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
130	U04		BALL	B06AC01	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL
140	U04		BALL	B06AC02	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
150	U04	VA5	BALL	B06AC05	2	3	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB LO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
160	U04	VA5	BALL	B06AC07	4	24	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0	VAQ	FOR TURN AROUND TIE-INS ONLY
170	U04	VA5	BALL	B06AC06	2	3	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB LO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
180	U04	VA5	BALL	B06AC08	4	16	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
190							***** RISING STEM BALL VALVES *****			VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
190											
200											
200		VA5	BALL	R06AC06	2	6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
210		VA5	BALL	R06AC08	8	12	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
220							***** BUTTERFLY VALVES *****				
220											
230											
230			BUTTERFLY	T06AC61	10	24	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
240							***** GATE VALVES *****				
240											
250											
250	U04		GATE	G06AC48	0.75	0.75	GATE 600# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0		
260	U04	817	GATE	G06AC42	12	12	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE GO	0	0		
270			GATE	W06AC70	3	4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
280			GATE	W06AC71	6	24	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
290							***** GLOBE VALVES *****				
290											
300											
300	U04		GLOBE	L06AC48	0.75	1.5	GLOBE 600# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
310			GLOBE	L06AC41	2	4	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
320			GLOBE	L06AC42	6	10	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
330							***** CHECK VALVES *****				
330											
340											
340	U04	VC1	CHECK	C06AC50	0.75	1	CHECK PISTON 600# RF LTCS A350 LF2 CL.1 TRIM 16	0	0		
350			CHECK	C06AC45	2	8	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
360			CHECK	C06AC52	10	24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16	0	0		
370							*****PIPE ABOVE GROUND*****				
370											
380											
380			PIPE	5358512	0.75	1.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
390			PIPE	5358515	2	2	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0		
400			PIPE	5358514	3	3	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
410											
410			PIPE	9001211	4	4	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE	0	0		
420			PIPE	9001311	6	6	LINEPIPE SMLS WT=7.92MM API-5L X60 BE	0	0		
430			PIPE	9001493	8	8	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE	0	0		
440			PIPE	9001488	10	10	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE	0	0		
450			PIPE	9001495	8	12	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE	0	0		
460			PIPE	9001491	10	14	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE	0	0		
470			PIPE	9005244	16	16	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE	0	0		
480			PIPE	9005242	18	18	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE	0	0		
490	U04	821	PIPE	9001477	8	12	LINEPIPE SMLS WT = 18.26 MM API-5L X60 BE	0	0		
500			PIPE	9005243	20	20	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE	0	0		
510			PIPE	9005254	24	24	LINEPIPE SMLS WT= 20.62 MM API-5L X60 BE	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 20" AND ABOVE. SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS	12/16/2024	BY OZAC	U04	Page 1 of 7	600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
520														
520			PIPE	5358511	0.5	0.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0					
530														
530							*****PIPE BELOW GROUND*****							
540														
540			PIPE	9001254CTD	4	4	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0					
550			PIPE	9001311CTD	6	6	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED	0	0					
560			PIPE	9001493CTD	8	8	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0					
570			PIPE	9001488CTD	10	10	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED	0	0					
580			PIPE	9001495CTD	12	12	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE COATED	0	0					
590			PIPE	9001491CTD	14	14	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE COATED	0	0					
600			PIPE	9005244CTD	16	16	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE COATED	0	0					
610			PIPE	9005242CTD	18	24	LINEPIPE SMLS WT = 17.48 MM API-5L X60 BE COATED	0	0					
620			PIPE	9005243CTD	20	20	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE COATED	0	0					
630			PIPE	9005254CTD	24	24	LINEPIPE SMLS WT= 20.62MM API-5L X60 BE COATED	0	0					
640														
640							*****FITTINGS ABOVE & BELOW GROUND*****							
650														
650			ELBOW	5392816	0.75	2	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0					
660			ELBOW	5392600	3	3	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0					
670			ELBOW	9002190	4	4	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60	0	0					
680			ELBOW	9006478	6	6	ELL BW 90 DEG LR WT 7.92 MM A860 WPHY 60	0	0					
690			ELBOW	9006471	8	8	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60	0	0					
700			ELBOW	9006468	10	10	ELL BW 90 DEG LR WT 11.13 MM A860 WPHY 60	0	0					
710			ELBOW	9006479	12	12	ELL BW 90 DEG LR WT 12.70 MM A860 WPHY 60	0	0					
720			ELBOW	9006480	14	14	ELL BW 90 DEG LR WT 14.27 MM A860 WPHY 60	0	0					
730			ELBOW	9006469	16	16	ELL BW 90 DEG LR WT 15.88 MM A860 WPHY 60	0	0					
740			ELBOW	9006474	18	18	ELL BW 90 DEG LR WT 17.48 MM A860 WPHY 60-W	0	0					
750			ELBOW	9006475	20	20	ELL BW 90 DEG LR WT 19.05 MM A860 WPHY 60-W	0	0					
760			ELBOW	9006476	24	24	ELL BW 90 DEG LR WT 20.62 MM A860 WPHY 60-W	0	0					
770			ELBOW	5392822	0.75	2	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0					
780			ELBOW	5392606	3	3	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0					
790			ELBOW	9006571	4	4	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60	0	0					
800			ELBOW	9006588	6	6	ELL BW 45 DEG LR WT 7.92MM A860 WPHY 60	0	0					
810			ELBOW	9006583	8	8	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60	0	0					
820			ELBOW	9006569	10	10	ELL BW 45 DEG LR WT 11.13MM A860 WPHY 60	0	0					
830			ELBOW	9006590	12	12	ELL BW 45 DEG LR WT 12.70MM A860 WPHY 60	0	0					
840			ELBOW	9006595	14	14	ELL BW 45 DEG LR WT 14.27MM A860 WPHY 60	0	0					
850			ELBOW	9006581	16	16	ELL BW 45 DEG LR WT 15.88MM A860 WPHY 60	0	0					
860			ELBOW	9006582	18	18	ELL BW 45 DEG LR WT 17.48MM A860 WPHY 60	0	0					
870			ELBOW	9006572	20	20	ELL BW 45 DEG LR WT 19.05MM A860 WPHY 60-W	0	0					
880			ELBOW	9006578	24	24	ELL BW 45 DEG LR WT 20.62MM A860 WPHY 60-W	0	0					
890														
890			TEE	5392825	0.75	2	TEE BW SCH 160 LTCS A420 WPL6	0	0					
900			TEE	5392609	3	3	TEE BW SCH XS LTCS A420 WPL6	0	0					
910		NT5	TEE	9301609	4	4	TEE BW WT 6.35 MM LTCS A860 WPHY-60	0	0					
920		NT5	TEE	9301606	6	6	TEE BW WT 7.92 MM LTCS A860 WPHY-60	0	0					
930		NT5	TEE	9301612	8	8	TEE BW WT 9.53 MM LTCS A860 WPHY-60	0	0					
940		NT5	TEE	9006604	10	10	TEE BW WT 11.13MM A860 WPHY 60	0	0					
950		NT5	TEE	9006633	12	12	TEE BW WT 12.7 MM A860 WPHY 60	0	0					
960		NT5	TEE	9006486	14	14	TEE BW WT 14.27MM A860 WPHY 60	0	0					
970		NT5	TEE	9004200	16	16	TEE BW WT 15.88MM A860 WPHY 60	0	0					
980		NT5	TEE	9004201	18	18	TEE BW WT 17.48MM A860 WPHY 60	0	0					
990		NT5	TEE	9004212	20	20	TEE BW WT 19.05MM A860 WPHY 60-W	0	0					
1000		NT5	TEE	9004213	24	24	TEE BW WT 20.62MM A860 WPHY 60-W	0	0					
1010			TEE RED	5392821	0.75	1.5	TEE RED BW SCH 160 LTCS A420 WPL6	1	2					
1020			TEE RED	5392469	1.5	2	TEE RED BW SCH XS/160 LTCS A420 WPL6	3	3					
1030		NT5	TEE RED	9006608	2	3	TEE RED BW WT 6.35 MM / SCH XS LTCS A860 WPHY 60	4	4					
1040		NT5	TEE RED	9006657	3	3	TEE RED BW WT 7.92 MM/SCH XS LTCS A860 WPHY 60	6	6					
1050		NT5	TEE RED	9006017	4	4	TEE RED BW WT 7.92 MM/6.35 MM LTCS A860 WPHY 60	6	6					
1060		NT5	TEE RED	9006100	4	4	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60	8	8					
1070		NT5	TEE RED	9006612	4	4	TEE RED BW WT 11.13 MM/6.35 MM LTCS A860 WPHY 60	10	10					
1080		NT5	TEE RED	9006649	6	6	TEE RED BW WT 11.13 MM/7.92 MM LTCS A860 WPHY 60	10	10					
1090		NT5	TEE RED	9006022	6	6	TEE RED BW WT 9.53 MM/7.92 MM LTCS A860 WPHY 60	8	8					
1100		NT5	TEE RED	9006641	6	6	TEE RED BW WT 12.70 MM/7.92 MM LTCS A860 WPHY 60	12	12					
1110		NT5	TEE RED	9006642	6	6	TEE RED BW WT 14.27 MM/7.92 MM LTCS A860 WPHY 60	14	14					
1120		NT5	TEE RED	9006643	8	8	TEE RED BW WT 12.70 MM/9.53 MM LTCS A860 WPHY 60	12	12					
1130			TEE RED	9006650	6	6	TEE RED BW WT 15.88 MM/7.92 MM LTCS A860 WPHY 60	16	16					
1140			TEE RED	9006650	6	6	TEE RED BW WT 15.88 MM/7.92 MM LTCS A860 WPHY 60	16	16					
1150		NT5	TEE RED	9004207	8	8	TEE RED BW WT 15.88 MM/9.53 MM LTCS A860 WPHY 60	16	16					
1160		NT5	TEE RED	9006644	10	10	TEE RED BW WT 15.88 MM/11.13 MM LTCS A860 WPHY 60	16	16					
1170		NT5	TEE RED	9006646	12	12	TEE RED BW WT 15.88 MM/12.70 MM LTCS A860 WPHY 60	16	16					

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 20" AND ABOVE. SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS	12/16/2024	BY OZAC	U04	Page 2 of 7	600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1180		NT5	TEE RED	9006651	14	14	TEE RED BW WT 15.88 MM/14.27 MM LTCS A860 WPHY 60	16	16	
1190		NT5	TEE RED	9301690	8	8	TEE RED BW WT 17.48 MM/9.53MM LTCS A860 WPHY-60	18	18	
1200		NT5	TEE RED	9006629	12	12	TEE RED BW WT 17.48 MM/12.7 MM LTCS A860 WPHY 60-W	18	18	
1210		NT5	TEE RED	9006656	16	16	TEE RED BW WT 17.48 MM/15.88 MM LTCS A860 WPHY 60-W	18	18	
1220			TEE RED	9006202	8	8	TEE RED BW WT 19.05 MM/9.53 MM LTCS A860 WPHY 60-W	20	20	
1230		NT5	TEE RED	9006660	12	12	TEE RED BW WT 19.05 MM/12.70 MM LTCS A860 WPHY 60-W	20	20	
1240		NT5	TEE RED	9006648	16	16	TEE RED BW WT 19.05 MM/15.88 MM LTCS A860 WPHY 60-W	20	20	
1250		NT5	TEE RED	9004202	10	10	TEE RED BW WT 20.62 MM/11.13 MM LTCS A860 WPHY 60-W	24	24	
1260		NT5	TEE RED	9006647	12	12	TEE RED BW WT 20.62 MM/12.70 MM LTCS A860 WPHY 60-W	24	24	
1270		NT5	TEE RED	9004203	16	16	TEE RED BW WT 20.62 MM/15.88 MM LTCS A860 WPHY 60-W	24	24	
1280		NT5	TEE RED	9004204	18	18	TEE RED BW WT 20.62 MM/17.48 MM LTCS A860 WPHY 60-W	24	24	
1290		NT5	TEE RED	9004205	20	20	TEE RED BW WT 20.62 MM/19.05 MM LTCS A860 WPHY 60-W	24	24	
1300	U04	821		9006696	8	8	TEE RED BW WT 12.7 MM/14.27 MM LTCS A860 WPHY 60	12	12	
1310	U04	NT5	TEE RED	9006617	10	10	TEE RED BW WT 12.70 MM/11.13 MM LTCS A860 WPHY 60	12	12	
1320	U04	NT5	TEE RED	9006667	10	10	TEE RED BW WT 14.27 MM/11.13 MM LTCS A860 WPHY 60	14	14	
1330	U04	821	TEE RED	9006673	12	12	TEE RED BW WT 17.48 MM/18.26 MM LTCS A860 WPHY 60	18	18	
1340	U04	821	TEE RED	9006678	12	12	TEE RED BW WT 23.82 MM Taper bored to 17.48MM/21.43 MM LTCS A860 WPHY 60	18	18	
1350	U04	821	TEE RED	9006677	10	10	TEE RED BW WT 30.96 MM Taper Bored to 15.88MM /21.44MM LTCS A860 WPHY 60	16	16	
1360	U04	821	TEE RED	9006668	12	12	TEE RED BW WT 38.10 MM Taper Bored to 19.05MM /25.4 MM LTCS A860 WPHY 60	20	20	
1370	U04	821	TEE RED	9006675	12	12	TEE RED BW WT 50.01MM Taper Bored to 19.05MM /33.32MM LTCS A860 WPHY 60	20	20	
1380	U04	821	TEE RED	9006676	16	16	TEE RED BW WT 59.54MM Taper Bored to 20.62 MM /40.49 MM LTCS A860 WPHY 60	24	24	
1390										
1400			CAP	5392826	0.75	2	CAP BW SCH 160 LTCS A420 WPL6	0	0	
1410			CAP	5392610	3	3	CAP BW SCH XS LTCS A420 WPL6	0	0	
1410			CAP	9005071	4	4	CAP BW WT 6.35 MM LTCS A860 WPHY-60	0	0	
1420			CAP	9005058	6	6	CAP BW WT 7.92 MM LTCS A860 WPHY-60	0	0	
1430			CAP	9005060	8	8	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0	
1440			CAP	9005064	10	10	CAP BW WT 11.13 MM LTCS A860 WPHY-60	0	0	
1450			CAP	9005067	12	12	CAP BW WT 12.70 MM LTCS A860 WPHY-60	0	0	
1460			CAP	9005068	14	14	CAP BW WT 14.27 MM LTCS A860 WPHY-60	0	0	
1470			CAP	9005069	16	16	CAP BW WT 15.88 MM LTCS A860 WPHY-60	0	0	
1480			CAP	9005070	18	18	CAP BW WT 17.48 MM LTCS A860 WPHY-60	0	0	
1490			CAP	9005072	20	20	CAP BW WT 19.05 MM LTCS A860 WPHY-60	0	0	
1500			CAP	9005073	24	24	CAP BW WT 20.62 MM LTCS A860 WPHY-60	0	0	
1510										
1510			RED CONC	5392827	0.75	1.5	REDUCER CONC SCH 160 LTCS A420 WPL6	1	2	
1520			RED ECC	5392828	0.75	1.5	REDUCER ECC SCH 160 LTCS A420 WPL6	1	2	
1530			RED CONC	5392462	1.5	2	REDUCER CONC SCH XS/160 LTCS A420 WPL6	3	3	
1540			RED ECC	5392465	1.5	2	REDUCER ECC SCH XS/160 LTCS A420 WPL6	3	3	
1550			RED CONC	9006923	1.5	2	REDUCER CONC WT 6.35MM / SCH 160 A860 WPHY 60	4	4	
1560			RED ECC	9002042	1.5	2	REDUCER ECC WT 6.35 MM / SCH 160 LTCS A860 WPHY-60	4	4	
1570			RED CONC	9006919	3	3	REDUCER CONC WT 6.35 MM / XS WT LTCS A860 WPHY-60	4	4	
1580			RED ECC	9001581	3	3	REDUCER ECC WT 6.35 MM / XS WT LTCS A860 WPHY-60	4	4	
1590			RED CONC	9006921	3	3	REDUCER CONC WT 7.92 MM / SCH XS MM LTCS A860 WPHY-60	6	6	
1600			RED ECC	9006922	3	3	REDUCER ECC WT 7.92 MM / SCH XS MM LTCS A860 WPHY-60	6	6	
1610			RED CONC	9006909	4	4	REDUCER CONC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	6	6	
1620			RED ECC	9002272	4	4	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	6	6	
1630			RED ECC	9002393	4	4	REDUCER ECC WT 9.53 MM / 6.35 MM LTCS A860 WPHY-60	8	8	
1640			RED ECC	9002251	6	6	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60	8	8	
1650			RED ECC	9001586	4	4	REDUCER ECC WT 11.13 MM / 6.35MM LTCS A860 WPHY-60	10	10	
1660			RED ECC	9002053	6	6	REDUCER ECC WT 11.13 MM / 7.92 MM LTCS A860 WPHY-60	10	10	
1670			RED ECC	9002403	8	8	REDUCER ECC WT 11.13 MM / 9.53 MM LTCS A860 WPHY-60	10	10	
1680			RED ECC	9002098	6	6	REDUCER ECC WT 12.70 MM / 7.92 MM LTCS A860 WPHY-60	12	12	
1690			RED ECC	9002405	8	8	REDUCER ECC WT 12.70 MM / 9.53 MM LTCS A860 WPHY-60	12	12	
1700			RED ECC	9002407	10	10	REDUCER ECC WT 12.70 MM / 11.13 MM LTCS A860 WPHY-60	12	12	
1710			RED ECC	9002025	8	8	REDUCER ECC WT 14.27 MM / 9.53 MM LTCS A860 WPHY-60	14	14	
1720			RED CONC	9006905	10	10	REDUCER CONC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60	14	14	
1730			RED ECC	9002409	4	10	REDUCER ECC WT 14.27 MM / 11.13 MM LTCS A860 WPHY-60	6	14	
1740			RED CONC	9006900	12	12	REDUCER CONC WT 14.27MM / 12.7MM LTCS A860 WPHY 60	14	14	
1750	U04		RED CONC	9006912	16	16	REDUCER CONC WT 20.62 MM / 15.88 MM LTCS A860 WPHY-60	24	24	
1760			RED ECC	9002410	12	12	REDUCER ECC WT 14.27 MM / 12.70 MM LTCS A860 WPHY-60	14	14	
1770			RED ECC	9004278	8	8	REDUCER ECC WT 15.88 MM / 9.53 MM LTCS A860 WPHY-60	16	16	
1780			RED ECC	9004279	10	10	REDUCER ECC WT 15.88 MM / 11.13 MM LTCS A860 WPHY-60	16	16	
1790			RED ECC	9004280	12	12	REDUCER ECC WT 15.88 MM / 12.70 MM LTCS A860 WPHY-60	16	16	
1800			RED ECC	9004299	14	14	REDUCER ECC WT 15.88 MM / 14.27MM LTCS A860 WPHY-60	16	16	
1810			RED ECC	9004282	10	10	REDUCER ECC WT 17.48 MM / 11.13 MM LTCS A860 WPHY-60	18	18	
1820			RED ECC	9004283	12	12	REDUCER ECC WT 17.48 MM / 12.70 MM LTCS A860 WPHY-60	18	18	
1830			RED ECC	9004300	14	14	REDUCER ECC WT 17.48 MM / 14.27MM LTCS A860 WPHY-60	18	18	
1840			RED CONC	9006906	16	16	REDUCER CONC WT 17.48 MM / 15.88MM LTCS A860 WPHY-60	18	18	
1850			RED ECC	9004284	16	16	REDUCER ECC WT 17.48 MM / 15.88MM LTCS A860 WPHY-60	18	18	
1860			RED ECC	9004302	16	16	REDUCER ECC WT 19.05MM / 15.88MM LTCS A860 WPHY-60-W	20	20	
1870			RED ECC	9004303	18	18	REDUCER ECC WT 19.05MM / 17.48 MM LTCS A860 WPHY-60-W	20	20	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 20" AND ABOVE. SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS	12/16/2024	BY OZAC	U04	Page 3 of 7	600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1880			RED CONC	9006913	16	16	REDUCER CONC WT 20.62 MM / 15.88 MM LTCS A860 WPHY-60-W	24	24	
1890			RED ECC	9002049	16	16	REDUCER ECC WT 20.62MM / 15.88MM LTCS A860 WPHY-60-W	24	24	
1900			RED CONC	9006911	18	18	REDUCER CONC WT 20.62 MM / 17.48 MM LTCS A860 WPHY-60-W	24	24	
1910			RED ECC	9001598	18	18	REDUCER ECC WT 20.62MM / 17.48 MM LTCS A860 WPHY-60-W	24	24	
1920			RED CONC	9009610	20	20	REDUCER CONC WT 20.62 MM / 19.05 MM LTCS A860 WPHY-60-W	24	24	
1930			RED ECC	9002050	20	20	REDUCER ECC WT 20.62MM / 19.05MM LTCS A860 WPHY-60-W	24	24	
1940	U04	821	RED ECC	9006505	6	6	REDUCER ECC WT 11.13MM / 14.27 MM LTCS A860 WPHY-60	10	10	
1950	U04	821	RED ECC	9001599	6	6	REDUCER ECC WT 12.7 MM / 14.27 MM LTCS A860 WPHY-60	12	12	
1960	U04		RED ECC	9004298	6	6	REDUCER ECC WT 14.27 MM / 7.92 MM LTCS A860 WPHY-60	14	14	
1970	U04	821		9002047	6	6	REDUCER ECC WT 14.27MM LTCS A860 WPHY-60	8	8	
1980	U04	821	RED ECC	9001384	8	8	REDUCER ECC WT 15.88 MM / 12.70MM LTCS A860 WPHY-60-W	12	12	
1990	U04	821		9001386	10	10	REDUCER ECC WT 15.88 MM / 14.27MM LTCS A860 WPHY-60-W	16	16	
2000	U04	821	RED ECC	9004312	8	8	REDUCER ECC WT 18.26MM LTCS A860 WPHY-60	12	12	
2010	U04		RED ECC	9002106	12	12	REDUCER ECC WT 19.05MM / 12.70MM LTCS A860 WPHY-60	20	20	
2020	U04		RED ECC	9004286	18	18	REDUCER ECC WT 19.05MM / 17.48 MM LTCS A860 WPHY-60	20	20	
2030	U04	821	RED ECC	9006508	6	6	REDUCER ECC WT 21.43MM / 14.27 MM LTCS A860 WPHY-60	10	10	
2040	U04	821	RED ECC	9006509	8	8	REDUCER ECC WT 21.43MM / 18.26 MM LTCS A860 WPHY-60	12	12	
2050	U04	821	RED ECC	9006503	8	8	REDUCER ECC WT 25.4MM / 18.26MM MM LTCS A860 WPHY-60	12	12	
2060	U04	821	RED ECC	9002425	8	8	REDUCER ECC WT 33.32 MM / 23.01 LTCS A860 WPHY-60	12	12	
2070	U04	821	RED ECC	9006507	10	10	REDUCER ECC WT 40.48MM / 28.57 MM LTCS A860 WPHY-60	16	16	
2080										
2080							***** INDUCTION BENDS ABOVE GROUND*****			
2090										
2090			INDUC BEND	9001534	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2100			INDUC BEND	9001815	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2110			INDUC BEND	9001817	8	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2120			INDUC BEND	9001899	10	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2130			INDUC BEND	9005321	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2140			INDUC BEND	9005322	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2150			INDUC BEND	9001398	16	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2160			INDUC BEND	9001573	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2170			INDUC BEND	9002060	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2180			INDUC BEND	9001574	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2190			INDUC BEND	9001575	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2200			INDUC BEND	9001825	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2210			INDUC BEND	9001827	8	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2220			INDUC BEND	9005326	10	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2230			INDUC BEND	9005327	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2240			INDUC BEND	9005328	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2250			INDUC BEND	9001200	16	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2260			INDUC BEND	9005040	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2270			INDUC BEND	9002061	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 19.05 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2280			INDUC BEND	9002026	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2290			INDUC BEND	9001580	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2300			INDUC BEND	9001835	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2310			INDUC BEND	9001837	8	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2320			INDUC BEND	9005331	10	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2330			INDUC BEND	9005333	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2340			INDUC BEND	9005334	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2350			INDUC BEND	9002002	16	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2360			INDUC BEND	9001579	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2370			INDUC BEND	9002062	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2380			INDUC BEND	9002027	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 20.62MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
2390										
2390							***** INDUCTION BENDS BELOW GROUND*****			
2400										
2400			INDUC BEND	9001534CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2410			INDUC BEND	9001815CTD	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2420			INDUC BEND	9001817CTD	8	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2430			INDUC BEND	9001899CTD	10	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2440			INDUC BEND	9005321CTD	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2450			INDUC BEND	9005322CTD	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2460			INDUC BEND	9001398CTD	16	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2470			INDUC BEND	9001573CTD	18	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2480			INDUC BEND	9002060CTD	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2490			INDUC BEND	9001574CTD	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 20" AND ABOVE. SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS	12/16/2024	BY OZAC	U04	Page 4 of 7	600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
2500	U04		INDUC BEND	9007040CTD	16	16	INDUCTION BEND 90 DEG X 10D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2510	U04		INDUC BEND	9007041CTD	18	18	INDUCTION BEND 90 DEG X 10D RADIUS WT 17.48MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2520	U04		INDUC BEND	9007036CTD	8	8	INDUCTION BEND 90 DEG X 10D RADIUS WT 9.53MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2530	U04		INDUC BEND	9007048CTD	16	16	INDUCTION BEND 90 DEG X 15D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2540	U04		INDUC BEND	9007046CTD	8	8	INDUCTION BEND 90 DEG X 15D RADIUS WT 9.53MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2550	U04		INDUC BEND	9007051CTD	8	8	INDUCTION BEND 90 DEG X 20D RADIUS WT 9.53MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2560	U04		INDUC BEND	9007052CTD	8	8	INDUCTION BEND 90 DEG X 25D RADIUS WT 9.53MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2570			INDUC BEND	9001575CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2580			INDUC BEND	9001825CTD	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2590			INDUC BEND	9001827CTD	8	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2600			INDUC BEND	9005326CTD	10	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2610			INDUC BEND	9005327CTD	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2620			INDUC BEND	9005328CTD	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2630			INDUC BEND	9001200CTD	16	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2640			INDUC BEND	9005040CTD	18	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2650			INDUC BEND	9002061CTD	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 19.05 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2660			INDUC BEND	9002026CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2670	U04		INDUC BEND	9007012CTD	16	16	INDUCTION BEND 45DEG X 10D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2680	U04		INDUC BEND	9007013CTD	18	18	INDUCTION BEND 45DEG X 10D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2690	U04		INDUC BEND	9007014CTD	24	24	INDUCTION BEND 45DEG X 10D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2700	U04		INDUC BEND	9007006CTD	8	8	INDUCTION BEND 45DEG X 10D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2710	U04		INDUC BEND	9007027CTD	16	16	INDUCTION BEND 45DEG X 15D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2720	U04		INDUC BEND	9007028CTD	18	18	INDUCTION BEND 45DEG X 15D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2730	U04		INDUC BEND	9007029CTD	24	24	INDUCTION BEND 45DEG X 15D RADIUS WT 20.62 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2740	U04		INDUC BEND	9007021CTD	8	8	INDUCTION BEND 45DEG X 15D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2750	U04		INDUC BEND	9007049CTD	16	16	INDUCTION BEND 45DEG X 20D RADIUS WT 15.88 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2760			INDUC BEND	9001580CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2770			INDUC BEND	9001835CTD	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2780			INDUC BEND	9001837CTD	8	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2790			INDUC BEND	9005331CTD	10	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2800			INDUC BEND	9004000CTD	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2810			INDUC BEND	9005334CTD	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 14.27 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2820			INDUC BEND	9002002CTD	16	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 15.88MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2830			INDUC BEND	9001579CTD	18	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 17.48 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2840			INDUC BEND	9002062CTD	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 19.05MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2850			INDUC BEND	9002027CTD	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 20.62MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0		
2860							*****OLETS ABOVE & BELOW GROUND*****				
2860											
2870											
2870	U04		OLETS	5505392	0.75	0.75	WELDOLET SCH 160 LTCS A350 LF2 CL.1	2	2		
2880	U04		OLETS	5505393	0.75	1	WELDOLET SCH XS/160 LTCS A350 LF2 CL.1	3	3		
2890			OLETS	9002450	0.75	1.5	WELDOLET WT 6.35 MM/SCH 160 WT LTCS A694 F60	4	4		
2900			OLETS	9006818	0.75	2	WELDOLET WT 7.92 MM/ SCH 160 WT LTCS A694 F60	6	6		
2910	U04		OLETS	9006823	0.75	2	WELDOLET WT 7.92 MM/ SCH 160 WT LTCS A694 F60	6	6		
2920			OLETS	9002457	0.75	2	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	8	8		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 20" AND ABOVE. SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS	12/16/2024	BY OZAC	U04	Page 5 of 7	600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
2930			OLETS	9006429	3	3	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	8	8		
2940			OLETS	9002076	0.75	2	WELDOLET WT 11.13MM/ SCH 160 LTCS A694 F60	10	10		
2950			OLETS	9005119	3	3	WELDOLET WT 11.13 MM/ SCH XS WT LTCS A694 F60	10	10		
2960			OLETS	9002079	4	4	WELDOLET WT 11.13MM/ 6.35MM LTCS A694 F60	10	10		
2970			OLETS	9002466	0.75	2	WELDOLET WT 12.70 MM/ SCH 160 WT LTCS A694 F60	12	12		
2980			OLETS	9002467	3	3	WELDOLET WT 12.70 MM/ SCH XS WT LTCS A694 F60	12	12		
2990			OLETS	9002469	4	4	WELDOLET WT 12.70 MM/ 6.35 MM WT LTCS A694 F60	12	12		
3000			OLETS	9002470	0.75	2	WELDOLET WT 14.27 MM/ SCH 160 WT LTCS A694 F60	14	14		
3010			OLETS	9002471	3	3	WELDOLET WT 14.27 MM/ SCH XS WT LTCS A694 F60	14	14		
3020			OLETS	9002473	4	4	WELDOLET WT 14.27 MM/ 6.35 MM LTCS A694 F60	14	14		
3030			OLETS	9002085	0.75	2	WELDOLET WT 15.88 MM/ SCH 160 LTCS A694 F60	16	16		
3040			OLETS	9005045	3	3	WELDOLET WT 15.88 MM/ SCH XS LTCS A694 F60	16	16		
3050			OLETS	9002088	4	4	WELDOLET WT 15.88 MM/ 6.35 MM LTCS A694 F60	16	16		
3060			OLETS	9004336	0.75	2	WELDOLET WT 17.48 MM/SCH 160 LTCS A694 F60	18	18		
3070			OLETS	9005053	3	3	WELDOLET WT 17.48 MM/SCH XS MM LTCS A694 F60	18	18		
3080			OLETS	9005049	4	4	WELDOLET WT 17.48 MM/6.35 MM LTCS A694 F60	18	18		
3090			OLETS	9005051	6	6	WELDOLET WT 17.48 MM/7.92 MM LTCS A694 F60	18	18		
3100			OLETS	9002089	0.75	2	WELDOLET WT 19.05 MM/ SCH 160 LTCS A694 F60	20	20		
3110			OLETS	9006252	3	3	WELDOLET WT 19.05 MM/ SCH XS LTCS A694 F60	20	20		
3120			OLETS	9002092	4	4	WELDOLET WT 19.05 MM/ 6.35 MM LTCS A694 F60	20	20		
3130			OLETS	9002094	6	6	WELDOLET WT 19.05 MM/ 7.92 MM LTCS A694 F60	20	20		
3140			OLETS	9004339	8	8	WELDOLET WT 19.05 MM/ 9.53 MM LTCS A694 F60	20	20		
3150			OLETS	9002096	0.75	2	WELDOLET WT 20.62 MM/ SCH 160 LTCS A694 F60	24	24		
3160			OLETS	9005126	3	3	WELDOLET WT 20.62 MM/ SCH XS LTCS A694 F60	24	24		
3170			OLETS	9002028	4	4	WELDOLET WT 20.62 MM/ 6.35 MM LTCS A694 F60	24	24		
3180			OLETS	9002824	6	6	WELDOLET WT 20.62 MM/ 7.92 MM LTCS A694 F60	24	24		
3190	U04		OLETS	9008204	1	1	WELDOLET FLAT BOTTOM SCH 160 LTCS A694 F60	0	0		
3200			OLETS	9005392	8	8	WELDOLET WT 20.62 MM/ 9.53 MM LTCS A694 F60	24	24		
3210											
3210	U04		OLETS	5504460	0.75	1	NIPOFLANGE 600# RF SCH 160 LTCS A350 LF2 CL.1	2	2		
3220	U04		OLETS	5504461	0.75	2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2 CL.1	3	3		
3230			OLETS	9002070	0.75	2	NIPOFLANGE 600# RF WT 6.35MM/SCH 160 LTCS A694 F60	4	4		
3240			OLETS	9006803	0.75	2	NIPOFLANGE 600# RF WT 7.92 MM/SCH 160 LTCS A694 F60	6	6		
3250			OLETS	9003455	0.75	2	NIPOFLANGE 600# RF 9.53 MM/SCH 160 WT LTCS A694 F60	8	8		
3260			OLETS	9003457	0.75	2	NIPOFLANGE 600# RF WT 11.13 MM/SCH 160 WT LTCS A694 F60	10	10		
3270			OLETS	9003458	0.75	2	NIPOFLANGE 600# RF WT 12.70 MM/SCH 160 WT LTCS A694 F60	12	12		
3280			OLETS	9003459	0.75	2	NIPOFLANGE 600# RF WT 14.27 MM/SCH 160 WT LTCS A694 F60	14	14		
3290			OLETS	9002067	0.75	2	NIPOFLANGE 600# RF WT 15.88MM/SCH 160 LTCS A694 F60	16	16		
3300			OLETS	9006496	0.75	2	NIPOFLANGE 600# RF 17.48 MM/SCH 160 LTCS A694 F60	18	18		
3310			OLETS	9002068	0.75	2	NIPOFLANGE 600# RF WT 19.05MM/SCH 160 LTCS A694 F60	20	20		
3320			OLETS	9002069	0.75	2	NIPOFLANGE 600# RF WT 20.62MM/SCH 160 LTCS A694 F60	24	24		
3330	U04	UJ2	OLETS	9003461	1.5	1.5	NIPOFLANGE 600# RF WT 15.88MM/SCH 160 LTCS A694 F60	16	16		
3340	U04	UJ2	OLETS	9003462	1	1	NIPOFLANGE 600# RF WT 19.05MM/SCH 160 LTCS A694 F60	20	20		
3350		FF5	OLETS	9504522	0.75	0.75	NIPOFLANGE FLAT BOTTOM 600# RF SCH 160 LTCS A694 F60	0	0		
3355	U04	UJ2	OLETS	9004322	1.5	1.5	NIPOFLANGE 1500# RF WT 15.88 MM/ SCH 160 LTCS A694 F60	16	16		
3360											
3360							***** FLANGES ABOVE & BELOW GROUND*****				
3370											
3370			FLANGE	5851204	0.5	2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE	0	0		
3380											
3380	U04		FLANGE	5557038	0.5	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 160	0	0		
3390	U04		FLANGE	5557036	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
3400			FLANGE	9001471	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	0	0		
3410			FLANGE	9003494	6	6	FLG WN 600# RF LTCS A694 F60 WT 7.92 MM	0	0		
3420			FLANGE	9003485	8	8	FLG WN 600# RF LTCS A694 F60 WT 9.53 MM	0	0		
3430			FLANGE	9001464	4	10	FLG WN 600# RF LTCS A694 F60 WT 11.13MM	0	0		
3440			FLANGE	9001512	12	12	FLG WN 600# RF LTCS A694 F60 WT 12.70MM	0	0		
3450			FLANGE	9001515	6	14	FLG WN 600# RF LTCS A694 F60 WT 14.27MM	0	0		
3460			FLANGE	9005128	16	16	FLG WN 600# RF LTCS A694 F60 WT 15.88 MM	0	0		
3470			FLANGE	9001513	18	18	FLG WN 600# RF LTCS A694 F60 WT 17.48MM	0	0		
3480	U04	821	FLANGE	9003439	8	8	FLG WN 600# RF LTCS A694 F60 WT 18.26 MM	0	0		
3490			FLANGE	9002064	20	20	FLG WN 600# RF LTCS A694 F60 WT 19.05 MM	0	0		
3500			FLANGE	9001517	24	24	FLG WN 600# RF LTCS A694 F60 WT 20.62MM	0	0		
3510	U04	821	FLANGE	9003440	10	10	FLG WN 600# RF LTCS A694 F60 WT 21.43 MM	0	0		
3520	U04	821	FLANGE	9005129	8	8	FLG WN 600# RF LTCS A694 F60 WT 23.01 MM	0	0		
3530	U04	821	FLANGE	9003479	10	10	FLG WN 600# RF LTCS A694 F60 WT 28.57 MM	0	0		
3540	U04	817	FLANGE	9006742	24	24	FLG WN 600# RTJ A694 F60 WT 20.62MM	0	0		
3550	U04	FS1	FLANGE	5557141	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
3560	U04	FS1	FLANGE	9002857	10	10	FLG WN 900# RF LTCS A694 F60 WT 11.13MM	0	0		
3570	U04	FS1	FLANGE	9006755	4	4	FLG WN 900# RF LTCS A694 F60 WT 6.35MM	0	0		
3580	U04	FS1	FLANGE	9004333	6	6	FLG WN 900# RF LTCS A694 F60 WT 7.92 MM	0	0		
3590	U04	FS1	FLANGE	9004328	8	8	FLG WN 900# RF LTCS A694 F60 WT 9.53 MM	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND. UNDER GROUND, PIPE SIZE 20" AND ABOVE. SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 93 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 95 °C COMMENT: LTCS	12/16/2024	BY OZAC	U04	Page 6 of 7	600K5H

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
3600	U04		BLIND	5615598	0.5	3	FLG BLIND 600# RF LTCS A350 LF2 CL.1	0	0	
3610			BLIND	9001518	4	24	FLG BLIND 600# RF LTCS A694 F60	0	0	
3620	U04		TEE RED	5599705	0.5	0.5	FLG SCR D (FNPT) REDUCING 600# RF A350 LF2 CL.1	0.75	1.5	
3630	U04	817	FLANGE	5558038	1	2	FLG WN 600# RTJ LTCS A350 LF2 CL.1 SCH 160	0	0	
3640	U04	817	FLANGE	9006704	4	4	FLG WN 600# RTJ A694 F60 WT 6.35MM	0	0	
3650	U04	817	FLANGE	9006704	4	4	FLG WN 600# RTJ A694 F60 WT 6.35MM	0	0	
3660	U04	817	FLANGE	9006711	6	6	FLG WN 600# RTJ A694 F60 WT 7.92MM	0	0	
3670	U04	817	FLANGE	9006723	8	8	FLG WN 600# RTJ A694 F60 WT 9.53MM	0	0	
3680	U04	817	FLANGE	9006706	10	10	FLG WN 600# RTJ A694 F60 WT 11.13MM	0	0	
3690	U04	817	FLANGE	9006707	12	12	FLG WN 600# RTJ A694 F60 WT 12.70MM	0	0	
3700	U04	817	FLANGE	9006714	14	14	FLG WN 600# RTJ A694 F60 WT 14.27MM	0	0	
3710	U04	817	FLANGE	9006712	16	16	FLG WN 600# RTJ A694 F60 WT 15.88MM	0	0	
3720	U04	817	FLANGE	9006709	18	18	FLG WN 600# RTJ A694 F60 WT 17.48MM	0	0	
3730	U04	817	FLANGE	9006713	20	20	FLG WN 600# RTJ A694 F60 WT 19.05MM	0	0	
3740	U04	817	BLIND	5615698	0.75	3	FLG BLIND 600# RTJ LTCS A350 LF2 CL.1	0	0	
3750	U04	817	BLIND	9006710	4	24	FLG BLIND 600# RTJ LTCS A694 F60	0	0	
3760	U04	817		5615616	2	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
3770	U04	817	BLIND	9002863	4	6	FLG BLIND 900# RF LTCS A694 F60	0	0	
3780	U04	817	FLANGE	5557281	2	3	FLG WN 900# RTJ LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
3790	U04	817	FLANGE	9006717	4	4	FLG WN 900# RTJ LTCS A694 F60 WT 6.35MM	0	0	
3800	U04	817	FLANGE	9006756	6	6	FLG WN 900# RTJ LTCS A694 F60 WT 7.92MM	0	0	
3810	U04	817	FLANGE	9006719	12	12	FLG WN 900# RTJ LTCS A694 F60 WT 12.70MM	0	0	
3820	U04	FS1	FLANGE	5557397	1	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
3830										
3830							***** GASKETS *****			
3840										
3840			GASKET	5673103	0.5	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3850	U04		GASKET	5672251	1	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3860	U04		GASKET	5673117	3	10	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
3870	U04	817	GASKET	5673202	0.75	24	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
3880	U04	817	GASKET	5674491	2	20	GASKET 900# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
3890										
3890		B1					***** BOLTING *****			
3900										
3900	U04		BOLTING	5675770	0.5	24	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
3910	U04	BA6	BOLTING	5675669	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M NUTS	0	0	
3920	U04	821		5675776	0.5	24	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
3930										
3930		SB6	BOLTING	5675675	12	24	JACK SCREW HEX HEAD A320 GR L7M	0	0	
3940										
3940			SPEC BLIND	5749587	0.75	8	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
3950										
3950							***** BLINDS & SPACERS *****			
3960										
3960			PADDLE	5750040	10	24	PADDLE BLANK 600# RF LTCS A516-GR70	0	0	
3970			PADDLE	5750251	10	24	PADDLE SPACER 600# RF LTCS A516-GR70	0	0	
3980										
3980	U04	817	SPEC BLIND	5749591	4	4	SPECTACLE BLIND 900# RTJ FEMALE LTCS A516-GR70	0	0	
3990										
3990	U04	817	PADDLE	5751040	12	12	PADDLE BLANK 600# RTJ LTCS A516-GR70	0	0	
4000	U04	817	PADDLE	5751251	12	12	PADDLE SPACER 600# RTJ LTCS A516-GR70	0	0	
4010										
4010							***** SP ITEMS *****			
4020										
4020			TEE	SP-9261	16	16	TEE BARRED EQUAL NPS 16 406.4MM OD X 15.88MM WT A860 WPHY 60	0	0	
4030										
4040										

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
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